# Linux Compression & Archiving Tools Notes

Linux provides multiple tools for **compression** (reducing file size) and **archiving** (bundling files). Below are detailed notes with examples.

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## 🔹 Compression Tools

### 1. gzip / gunzip (used for files)

- **Extension**: `.gz`
- **Commands:**

```
gzip file.txt        # compress → file.txt.gz (original deleted)
gunzip file.txt.gz   # decompress → restores file.txt
```

- **Keep original while compressing:**

```
gzip -k file.txt                    # creates file.txt.gz and keeps
file.txt
gzip -c file.txt > file.txt.gz      # compresses and redirects output
```

📌 **Difference between** `-k` **and** `-c`**:** - `-k` → gzip itself keeps original file. - `-c >` → write compressed data to stdout, redirect manually.

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### 2. bzip2 / bunzip2 (used for files)

- **Extension**: `.bz2`
- **Commands:**

```
bzip2 file.txt        # compress → file.txt.bz2 (original deleted)
bunzip2 file.txt.bz2  # decompress → restores file.txt
```

- **Keep original while compressing:**

```
bzip2 -k file.txt
bzip2 -c file.txt > file.txt.bz2
```

📌**Note**: Slower than gzip, but better compression.

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### 3. xz / unxz (used for files)

- **Extension**: `.xz`
- **Commands:**

```
xz file.txt         # compress → file.txt.xz (original deleted)
unxz file.txt.xz    # decompress → restores file.txt
```

- **Keep original:**

```
xz -k file.txt
```

📌**Note**: Highest compression ratio but slowest.

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### 4. zip / unzip (used for files and directories)

- **Extension**: `.zip`
- **Commands:**

```
zip archive.zip file1 file2     # compress files
unzip archive.zip               # extract files
```

- **Compress a directory:**

```
zip -r archive.zip mydir/
```

- **List contents of zip:**

```
unzip -l archive.zip
```

📌**Pros:** Cross-platform (Linux, Windows, macOS). 📌**Cons:** Compression ratio lower than xz/bzip2.

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## 🔹 Comparison Table

| Tool | Compress Command | Decompress Command | Extension | Use Case |
|---|---|---|---|---|
| gzip | `gzip file.txt` | `gunzip file.txt.gz` | .gz | Fast, common compression |
| bzip2 | `bzip2 file.txt` | `bunzip2 file.txt.bz2` | .bz2 | Better compression, slower |
| xz | `xz file.txt` | `unxz file.txt.xz` | .xz | Best compression, slowest |
| zip | `zip archive.zip file1 file2` | `unzip archive.zip` | .zip | Cross-platform archives |

📌**Note:** gzip, bzip2, and xz only compress **one file** at a time. For multiple files, use **tar + compression**.

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# ♒ Tape Archive (`tar`)

`tar` is an **archiving tool** that bundles multiple files/folders into one file (called a *tarball*). It can optionally compress using gzip, bzip2, or xz.

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## 🔹 Common Options

| Option | Meaning |
|---|---|
| `-c` | Create archive |
| `-x` | Extract archive |
| `-t` | List contents |
| `-f` | File name |
| `-v` | Verbose (show files as they are processed) |
| `-z` | Use gzip compression |
| `-j` | Use bzip2 compression |
| `-J` | Use xz compression |

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## 🔹 Creating Archives

1. **Plain tar (no compression)**

```
tar -cvf archive.tar file1.txt file2.txt dir/
```

🔔Creates `archive.tar` containing the files.

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1. **With gzip compression**

```
tar -czvf archive.tar.gz file1.txt file2.txt dir/
```

🔔Creates `archive.tar.gz` (fast, common).

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1. **With bzip2 compression**

```
tar -cjvf archive.tar.bz2 file1.txt file2.txt dir/
```

🔔Creates `archive.tar.bz2` (better compression, slower).

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1. **With xz compression**

```
tar -cJvf archive.tar.xz file1.txt file2.txt dir/
```

🔔Creates `archive.tar.xz` (best compression ratio).

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## 🔹 Extracting Archives

1. **Plain tar**

```
tar -xvf archive.tar
```

2. **.tar.gz**

```
tar -xzvf archive.tar.gz
```

3. **.tar.bz2**

```
tar -xjvf archive.tar.bz2
```

4. **.tar.xz**

```
tar -xJvf archive.tar.xz
```

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## 🔹 Listing Contents

```
tar -tvf archive.tar
```

🔔Shows list of files inside without extracting.

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## 🔹 Extract to Specific Directory

```
tar -xzvf archive.tar.gz -C /tmp/myfolder
```

🔔Extracts into `/tmp/myfolder`.

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## 🔹 Summary of Formats

| Format | Command Example | Extension | Best For |
|---|---|---|---|
| Plain tar | `tar -cvf archive.tar` | .tar | Archiving only |
| Gzip tar | `tar -czvf archive.tar.gz` | .tar.gz | Fast, common |
| Bzip2 tar | `tar -cjvf archive.tar.bz2` | .tar.bz2 | Better compression |
| XZ tar | `tar -cJvf archive.tar.xz` | .tar.xz | Best compression |

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## 🔹 Why Compress with tar?

- **tar** alone just bundles files.
- Compression (gzip, bzip2, xz) makes the archive smaller.
- Different methods offer trade-offs between **speed** and **compression ratio**.

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## 🔹 Example Workflow

### Compress multiple files into one compressed archive

```
tar -cvf archive.tar file1.txt file2.txt dir/
gzip archive.tar
```

🔔Creates `archive.tar.gz`.

Or directly:

```
tar -czvf archive.tar.gz file1.txt file2.txt dir/
```

🔔Creates compressed tarball in one step.

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🔗**In short:** - Use `gzip`, `bzip2`, `xz` for single files. - Use `tar` for multiple files/folders. - Combine `tar + compression` for efficient backups. - Use `zip` for cross-platform sharing.